

OMINATO AB, Japan

WMO: 475160

Lat: 41.233N

Lon: 141.132E

Elev: 10

StdP: 101.20

Time Zone: 9.00 (E09)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.5	(c) -6.2	(d) -12.9	(e) 1.2	(f) -5.3	(g) -11.2	(h) 1.4	(i) -3.8	(j) 11.9	(k) 0.1	(l) 10.5	(m) -0.4	(n) 3.1	(o) 290	(p) 0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.8	(c) 29.1	(d) 24.0	(e) 27.2	(f) 22.9	(g) 26.0	(h) 22.3	(i) 24.7	(j) 28.1	(k) 23.8	(l) 26.6	(m) 23.0	(n) 25.4	(o) 4.6	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.5	27.0	22.9	17.7	26.1	22.1	16.8	24.9	74.9	28.4	71.2	26.9	68.2	25.2	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.8	9.4	8.3	-11.0	31.1	2.5	1.9	-12.8	32.4	-14.3	33.6	-15.7	34.6	-17.5	36.0
			-11.5	25.5	2.4	0.9	-13.2	26.2	-14.6	26.7	-15.9	27.3	-17.6	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-1.5	-1.1	1.5	7.3	12.5	16.5	20.4	22.4	19.0	13.1	6.8	1.1
	DBStd	8.79	2.68	2.80	2.80	3.35	2.81	2.82	3.32	2.70	2.93	2.94	3.52	2.94
	HDD10.0	1419	355	309	263	94	9	0	0	0	0	7	105	276
	HDD18.3	3351	613	543	522	332	183	68	18	2	26	164	346	534
	CDD10.0	1380	0	0	0	13	85	195	322	383	270	102	10	0
	CDD18.3	270	0	0	0	1	1	13	82	127	46	1	0	0
	CDH23.3	1168	0	0	0	3	4	48	344	611	158	1	0	0
	CDH26.7	214	0	0	0	1	0	4	67	124	19	0	0	0
(14) Wind	WSAvg	3.8	3.9	4.0	4.4	4.4	4.2	3.7	3.6	3.3	3.4	3.5	3.7	3.9
(15) Precipitation	PrecAvg	1320	98	80	77	76	95	91	134	150	168	113	119	106
	PrecMax	1665	201	156	119	169	167	245	353	301	320	222	225	186
	PrecMin	984	46	36	36	12	32	19	21	41	38	38	53	37
	PrecStd	158	37	34	25	34	36	49	68	66	71	49	50	38
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	7.4	12.2	20.8	24.1	26.9	31.1	31.5	28.9	22.0	17.2	11.2
		MCWB	5.5	4.7	8.1	13.1	15.1	19.4	24.3	25.2	23.8	17.1	13.5	8.4
	2%	DB	5.0	5.8	9.1	17.0	21.8	24.8	28.9	29.8	26.8	20.2	15.8	9.1
		MCWB	2.8	3.3	5.9	10.9	14.7	18.9	23.7	24.6	22.7	16.3	12.7	7.1
	5%	DB	3.9	4.4	7.8	14.8	19.2	23.0	27.0	28.0	25.1	19.2	14.0	7.3
		MCWB	1.9	2.1	5.2	9.7	13.9	18.0	22.7	23.7	21.7	15.8	11.0	5.0
	10%	DB	2.8	3.3	6.0	12.9	17.3	21.2	25.2	26.2	23.8	18.2	12.8	5.9
		MCWB	0.9	1.1	3.3	8.7	12.9	17.5	21.8	22.9	20.6	14.9	10.3	3.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	5.3	8.9	14.1	17.1	21.0	25.2	26.0	24.8	18.8	14.8	9.4
		MCDB	6.5	6.6	11.7	18.5	21.3	24.7	29.7	30.0	27.6	20.3	16.5	10.8
	2%	WB	3.2	3.6	6.7	11.9	15.6	19.8	24.1	25.1	23.6	17.4	13.0	7.3
		MCDB	4.3	5.1	8.5	15.5	20.0	22.8	27.7	28.7	25.7	19.5	15.0	8.7
	5%	WB	2.0	2.5	5.0	10.5	14.5	19.0	23.3	24.3	22.5	16.5	11.8	5.3
		MCDB	3.3	4.1	7.0	13.6	18.2	22.0	26.3	27.3	24.5	18.7	13.6	7.0
	10%	WB	1.0	1.5	3.8	9.3	13.6	18.0	22.3	23.5	21.1	15.6	10.6	3.8
		MCDB	2.3	3.1	5.6	12.2	16.6	20.9	24.8	25.9	23.1	17.8	12.3	5.5
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	5.6	6.3	8.0	7.8	7.2	5.7	5.8	6.9	8.1	7.0	5.4
		MCDBR	6.4	6.8	8.8	11.5	11.3	10.2	8.2	8.0	7.9	8.7	9.3	7.6
	5% WB	MCWBR	5.6	5.5	6.9	7.0	6.4	5.7	4.4	4.3	4.8	5.8	7.2	6.5
		MCWBR	6.3	6.7	8.3	9.9	9.6	8.5	7.5	7.1	7.1	7.2	8.5	7.4
(40) Clear-Sky Solar Irradiance	taub		0.366	0.409	0.481	0.534	0.550	0.523	0.523	0.492	0.434	0.413	0.397	0.381
		taud	2.108	1.992	1.873	1.805	1.854	2.006	2.080	2.180	2.297	2.218	2.150	2.096
	Ebn at Noon		745	767	754	747	748	770	765	778	800	763	705	687
		Edh at Noon	109	142	180	207	202	174	160	141	117	113	104	102
(44) All-Sky Solar Radiation	RadAvg		1.41	2.17	3.34	4.49	4.94	4.86	4.29	4.15	3.66	2.79	1.68	1.19
	RadStd		0.08	0.19	0.23	0.46	0.49	0.39	0.45	0.44	0.32	0.22	0.14	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page